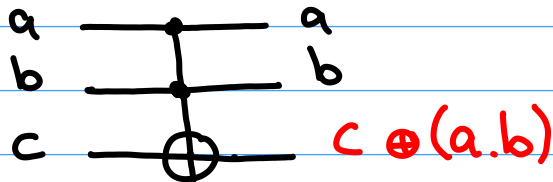
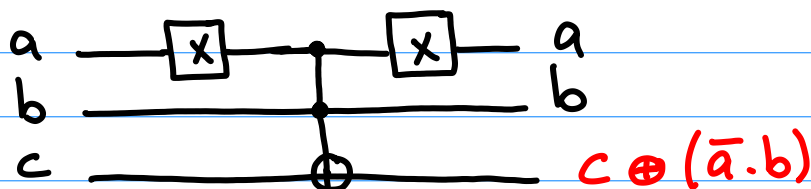


Quantum Algorithms:

Toffoli Gate:



if $ab = 11$ then $X|c\rangle$

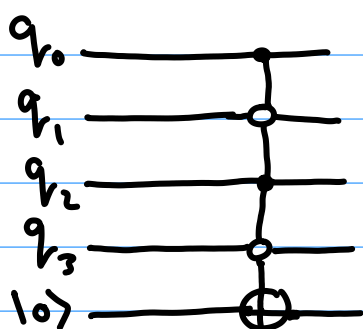


if $ab = 01$ then $X|c\rangle$

binary representation

2 qubits $\{0,1\}^2 \Rightarrow \{00, 01, 11, 11\}$ 4 inputs

n qubits $\{0,1\}^n \Rightarrow \{00\dots0, \dots, 111\dots1\}$
 2^n inputs



input $10 \Rightarrow 1010$

$q_0 q_1 q_2 q_3 = 1010$ then
 $X|0\rangle = |1\rangle$

Oracle:

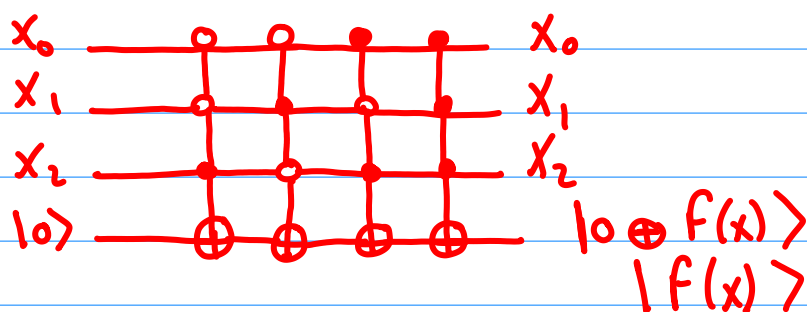
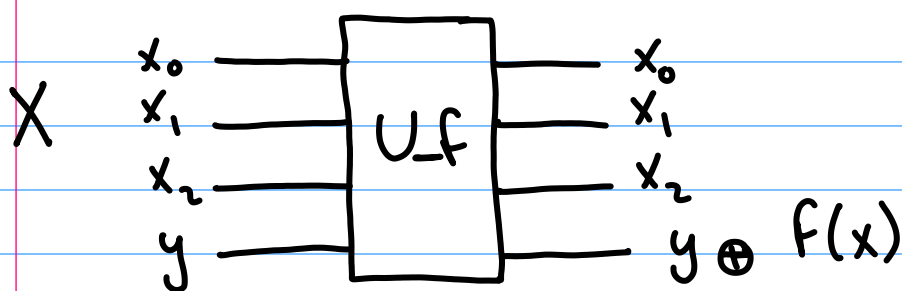
A function and treated as a black box

$$f(x) = y \quad x: \{0,1\}^n \quad y: \{0,1\}$$

in quantum circuits:

oracle is defined by unitary U_f

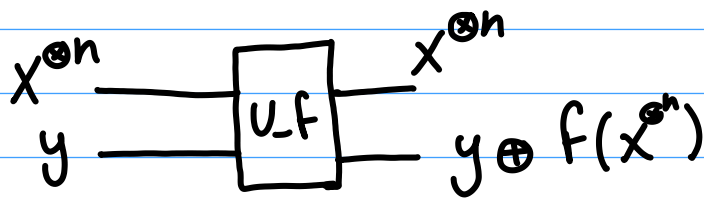
$$U_f |x\rangle |y\rangle = |x\rangle |y \oplus f(x)\rangle$$



eg. $f(x)$

x_0	x_1	x_2	$f(x)$
0	0	0	0
0	0	1	1
0	1	0	1
0	1	1	0
1	0	0	0
1	0	1	1
1	1	0	0
1	1	1	1

General: it is called query $f(x)$



phase Kickback:

A is a matrix

b is a vector

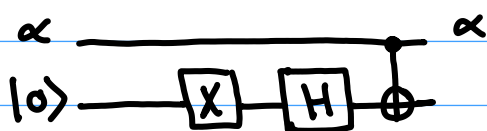
$$Ab = \lambda b$$

eigen vector eigen Value

$$|\psi\rangle = \frac{1}{\sqrt{2}} |0\rangle - \frac{1}{\sqrt{2}} |1\rangle = |-\rangle$$

$$\begin{aligned} X|\psi\rangle &= \frac{1}{\sqrt{2}} |1\rangle - \frac{1}{\sqrt{2}} |0\rangle \\ &= -\left(\frac{1}{\sqrt{2}} |0\rangle - \frac{1}{\sqrt{2}} |1\rangle\right) = -|-\rangle \end{aligned}$$

$X|-\rangle = -|-\rangle$
eigen vector eigen Value



$$\alpha = \{0, 1\}$$

$|\alpha\rangle|0\rangle$ apply X
 $|\alpha\rangle|1\rangle$ apply H

$$|\alpha\rangle|-\rangle = \frac{1}{\sqrt{2}} |\alpha\rangle|0\rangle - \frac{1}{\sqrt{2}} |\alpha\rangle|1\rangle$$

apply CNOT

$$\frac{1}{\sqrt{2}} |\alpha\rangle|\alpha\rangle - \frac{1}{\sqrt{2}} |\alpha\rangle|\bar{\alpha}\rangle$$

$$|\alpha\rangle \left(\frac{1}{\sqrt{2}} |\alpha\rangle - \frac{1}{\sqrt{2}} |\bar{\alpha}\rangle \right)$$

if $\alpha = 0$

$$|0\rangle \left(\frac{1}{\sqrt{2}} |0\rangle - \frac{1}{\sqrt{2}} |1\rangle \right) = |0\rangle|-\rangle$$

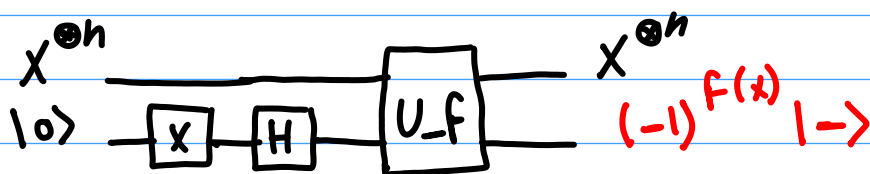
if $\alpha = 1$

$$|1\rangle \left(\frac{1}{\sqrt{2}} |1\rangle - \frac{1}{\sqrt{2}} |0\rangle \right) = -|1\rangle|-\rangle$$

↑
phase Kickback

When the condition is met apply phase to the state.

general:



$$U_f |x\rangle^{\oplus n} |-\rangle = (-1)^{f(x)} |x\rangle^{\oplus n} |-\rangle$$

This is the engine for DJ and BV

Deutsch Algorithm:

problem statement:

$f(x)$ $x: \{0,1\}$ $y: \{0,1\}$

↙ ↘
black box oracle (problem promise)

$f(x)$ could be constant function or balanced function

const.

	x	y	or	x	y
$f(x) = 1$	0	0		0	1
$f(x) = 0$	1	0		1	1

balanced

Half the output is 0 and the other half is 1 for all x

$f(x)$ is balanced

x	y	or	x	y
0	0		0	1
1	1		1	0

So we have $f(x)$ which is constant or balanced. We can input as many queries as we want. We want to determine if the $f(x)$

const. balanced

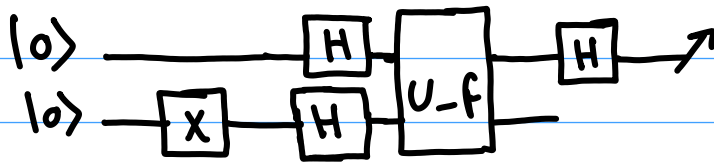
in classical solution:

$O(2)$ We need two queries

Try $x = 0$ $y = 0$ $\left\{ \begin{array}{l} \rightarrow \text{could be all 0's} \\ \rightarrow \text{could be balanced} \end{array} \right.$

We need a second query to determine

Quantum solution:



$$|0\rangle|0\rangle$$

X on q_1

$$|0\rangle|1\rangle$$

H on q_1

$$|0\rangle|-\rangle$$

H on q_0

$$\left(\frac{1}{\sqrt{2}} |0\rangle + \frac{1}{\sqrt{2}} |1\rangle \right) \otimes |-\rangle \quad \text{apply } U-f$$

$$\frac{1}{\sqrt{2}} U-f |0\rangle |-\rangle + \frac{1}{\sqrt{2}} U-f |1\rangle |-\rangle$$

$$\left[\frac{1}{\sqrt{2}} (-1)^{f(0)} |0\rangle + \frac{1}{\sqrt{2}} (-1)^{f(1)} |1\rangle \right] \otimes |-\rangle$$

$$(-1)^{f(0)} \left[\frac{|0\rangle + (-1)^{f(1)-f(0)} |1\rangle}{\sqrt{2}} \right] \otimes |-\rangle$$

apply H q_0

Note:

$$H \frac{|0\rangle + (-1)^K |1\rangle}{\sqrt{2}} = |K\rangle$$



$$K \bmod 2$$

$$K: \{0, 1\}$$

$$f(1) - f(0) \pmod{2}$$

$$f(1) - f(0) \pmod{2} \equiv f(1) \oplus f(0)$$

$$\begin{array}{ccc} 0 & 0 & 0 \\ 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{array} \equiv \begin{array}{c} 0 \\ 1 \\ 1 \\ 0 \end{array}$$

$$\underbrace{(-1)^{f(0)}}_{\downarrow} |f(1) \oplus f(0)\rangle \rightarrow \text{apply } M \text{ on } q_0$$

phase ignored

$$f(1) \oplus f(0) \Rightarrow \text{output on } q_0$$

$$\text{if } f(x) \text{ is constant } f(0) = f(1)$$

$$f(1) \oplus f(0) = |0\rangle$$

$$\text{if } f(x) \text{ is balanced } f(0) \neq f(1)$$

$$f(1) \oplus f(0) = |1\rangle$$

so after measurement

$$\begin{array}{ll} C_0 = |0\rangle & \text{const.} \\ C_0 = |1\rangle & \text{balanced} \end{array}$$

So Deutsch algorithm determines if $f(x)$ in just one query $O(1)$ instead of $O(2)$

Deutsch-Jozsa:

scale the problem $f: \{0,1\}^n \rightarrow \{0,1\}$

the same promise $f(x)$

constant balanced

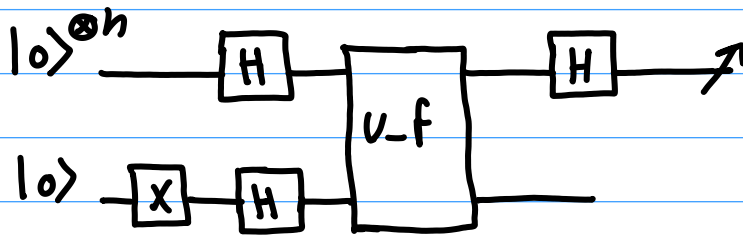
00...0	0	OR	1	Half 0s
00...1	0		1	Half 1s
⋮	⋮		⋮	
11...1	0		1	

classical solution:

We need $2^{n-1} + 1$ queries to determine $f(x)$
 $O(2^n)$ exponential

Quantum Solution:

$$\begin{array}{c} x_0 \\ x_1 \\ \vdots \\ x_n \end{array} \equiv x^{\otimes n}$$



$$|0\rangle^{\otimes n} |0\rangle$$

X on q_{n+1}

$$|0\rangle^{\otimes n} |1\rangle$$

H on q_{n+1}

$$|0\rangle^{\otimes n} |-\rangle$$

$H^{\otimes n}$ on $|0\rangle^{\otimes n}$

Note:

$$H^{\otimes n} |0\rangle^{\otimes n} = \frac{1}{\sqrt{2^n}} \sum_{x=0}^{2^n-1} |x\rangle$$

$$\text{eg. } H^{\otimes 2} |00\rangle = \frac{1}{\sqrt{2^2}} \sum_{x=0}^3 |x\rangle$$

$$= \frac{1}{2} (|00\rangle + |01\rangle + |10\rangle + |11\rangle)$$

$$H^{\otimes n} |0\rangle^{\otimes n} \mapsto$$

$$\frac{1}{\sqrt{2^n}} \left[\sum_{x=0}^{2^n-1} |x\rangle \right] \otimes |-\rangle$$

apply U-f

$$\frac{1}{\sqrt{2^n}} \left[\sum_{x=0}^{2^n-1} U-f |x\rangle \right] \otimes |-\rangle$$

$$\frac{1}{\sqrt{2^n}} \left[\sum_{x=0}^{2^n-1} (-1)^{f(x)} |x\rangle \right] \otimes |-\rangle$$

apply $H^{\otimes n}$

Note:

$$H^{\otimes n} |y\rangle = \frac{1}{\sqrt{2^n}} \sum_{x=0}^{2^n-1} (-1)^{x \cdot y} |x\rangle$$

$$y: \{0, 1\}^n$$

$$x \cdot y = x_0 y_0 \oplus x_1 y_1 \oplus \dots \oplus x_{n-1} y_{n-1}$$

eg.

$$H^{\otimes 3} |101\rangle = \frac{1}{\sqrt{2^3}} \sum_{x=0}^{2^3-1} (-1)^{x \cdot (101)} |x\rangle$$

$$000 \cdot 101 = 0 \oplus 0 \oplus 0 = 0$$

$$\frac{1}{2\sqrt{2}} \left((-1)^0 |000\rangle + (-1)^1 |001\rangle + (-1)^0 |010\rangle + (-1)^1 |011\rangle + \right. \\ \left. (-1)^1 |100\rangle + (-1)^0 |101\rangle + (-1)^1 |110\rangle + (-1)^0 |111\rangle \right)$$

$$\begin{aligned} 101 \cdot 101 &= 1 \cdot 1 \oplus 0 \cdot 0 \oplus 1 \cdot 1 \\ &= 1 \oplus 0 \oplus 1 = 0 \end{aligned}$$

$$\frac{1}{\sqrt{2^n}} \left[\sum_{x=0}^{2^n-1} (-1)^{f(x)} |x\rangle \right] \otimes |-\rangle \quad \text{apply } H^{\otimes n}$$

$$\frac{1}{\sqrt{2^n}} \left[\sum_{x=0}^{2^n-1} (-1)^{f(x)} H^{\otimes n} |x\rangle \right] \otimes |-\rangle$$

$$\frac{1}{\sqrt{2^n}} \left[\sum_{x=0}^{2^n-1} (-1)^{f(x)} \frac{1}{\sqrt{2^n}} \sum_{y=0}^{2^n-1} (-1)^{y \cdot x} |y\rangle \right] \otimes |-\rangle$$

$$\frac{1}{2^n} \left[\sum_{x=0}^{2^n-1} \sum_{y=0}^{2^n-1} (-1)^{f(x)} (-1)^{y \cdot x} |y\rangle \right] \otimes |-\rangle \quad \uparrow \text{ignore}$$

$$\sum_{y=0}^{2^n-1} \left[\frac{1}{2^n} \sum_{x=0}^{2^n-1} (-1)^{f(x) \oplus y \cdot x} \right] |y\rangle$$

coefficient of $|y\rangle$

apply M on $q_0, \dots, q_n$

$p(00\dots0)$ Where $|y\rangle = |00\dots0\rangle$

coefficient of $|00\dots0\rangle \quad y \cdot x = 0$

$$\frac{1}{2^n} \sum_{x=0}^{2^n-1} (-1)^{f(x)}$$

if $f(x)$ is constant $f(x) = c$

$$\frac{1}{2^n} \sum_{x=0}^{2^n-1} (-1)^c$$

$$\frac{1}{2^n} \cdot 2^n \cdot (-1)^c = (-1)^c$$

amplitude of $|00\dots 0\rangle$ and $f(x)$ is constant

$$p(00\dots 0) = |(-1)^c|^2 = 1$$

so if we measure $q_0, q_1, \dots, q_n$ and got all zeros then $f(x)$ is constant

if $f(x)$ is balanced

$$\frac{1}{\sqrt{2^n}} \sum_{x=0}^{2^n-1} (-1)^{f(x)}$$

$$1 + (-1) + 1 + \dots + (-1)$$

$$\sum = 0$$

So if $f(x)$ is balanced then

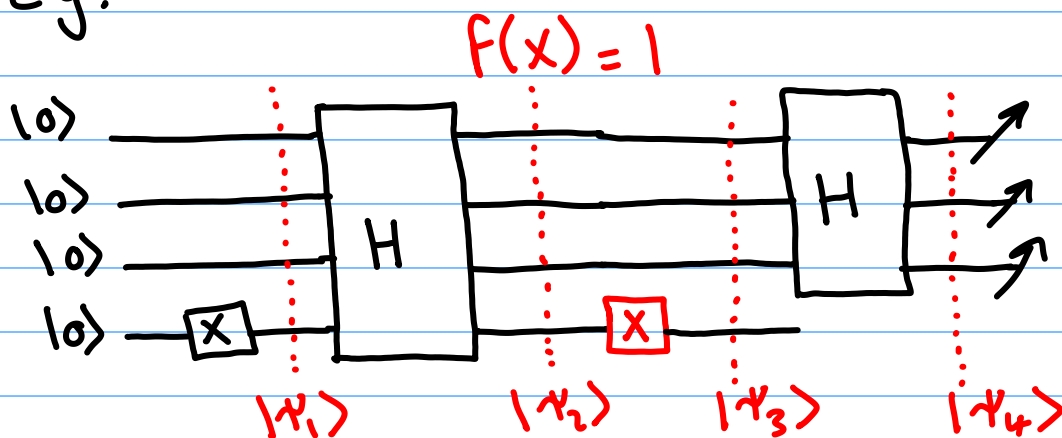
amplitude of $|y\rangle = |00\dots 0\rangle$ is 0

so $p(00\dots 0) = 0$

using Deutsch-Jozsa Algorithm, we can determine if $f(x)$ is constant or balanced

in just one query $O(1)$ achieving exponential speedup

eg.



$|000\rangle|0\rangle$

$|\psi_1\rangle = |000\rangle|1\rangle$

$|\psi_2\rangle = |+++ \rangle |-\rangle$

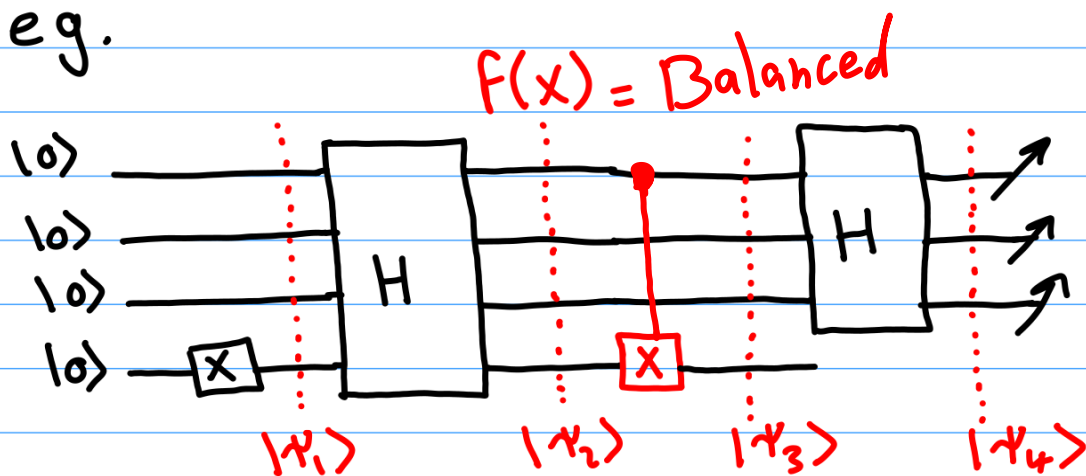
$$|\psi_3\rangle = -|+++ \rangle |-\rangle$$

$$|\psi_4\rangle = -|000 \rangle |-\rangle$$

After M

000 all zeros then $f(x)$ is constant

eg.



$$|000\rangle|0\rangle$$

$$|\psi_1\rangle = |000\rangle|1\rangle$$

$$|\psi_2\rangle = |+++ \rangle |-\rangle$$

0	0	0	0
0	0	1	0
0	1	0	0
0	1	1	0
1	0	0	1
1	0	1	1
1	1	0	1
1	1	1	1

$$|\psi_2\rangle = \frac{1}{2\sqrt{2}} \left(\begin{array}{l} |000\rangle|-\rangle + |001\rangle|-\rangle + |010\rangle|-\rangle + |011\rangle|-\rangle + \\ |100\rangle|-\rangle + |101\rangle|-\rangle + |110\rangle|-\rangle + |111\rangle|-\rangle \end{array} \right)$$

$$|\psi_3\rangle = \frac{1}{2\sqrt{2}} \left(\begin{array}{l} |000\rangle + |001\rangle + |010\rangle + |011\rangle \\ -|100\rangle - |101\rangle - |110\rangle - |111\rangle \end{array} \right) \otimes |-\rangle$$

$$|\psi_3\rangle = \frac{1}{2} (|-\rangle|00\rangle + |-\rangle|01\rangle + |-\rangle|10\rangle + |-\rangle|11\rangle) \otimes |-\rangle$$

$$|\psi_3\rangle = |---\rangle |-\rangle$$

$$|\psi_4\rangle = |100\rangle |-\rangle$$

after M 100 so $f(x)$ is Balanced

Bernstein-Vazirani:

problem statement:

We have $f(x)$ hides a secret string

$$f(x) = S \cdot X \pmod{2}$$

"pairwise product"

$$S = 101$$

0	0	0	0
0	0	1	1
0	1	0	0
0	1	1	1
1	0	0	1
1	0	1	0
1	1	0	1
1	1	1	0

$f(x)$ is a black box and
We need to find S

classical solution: $S = S_0 S_1 S_2$

$$X = 100 \quad f(100) = S \cdot 100$$

$$= S_0 \cdot 1 \oplus S_1 \cdot 0 \oplus S_2 \cdot 0$$

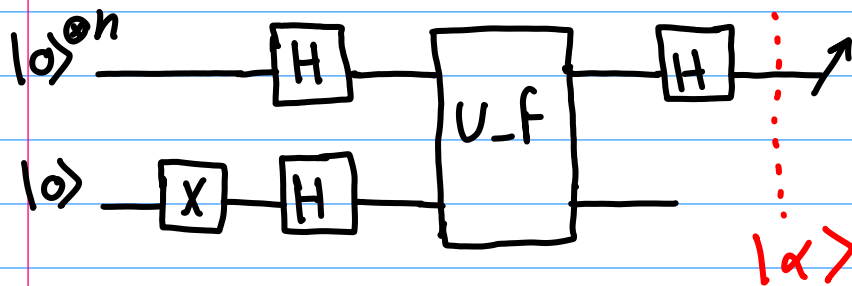
$$= S_0 \oplus 0 \oplus 0 = S_0$$

$$f(100) = S_0 = 1 \quad (\text{from the table})$$

do the same for $X=010$ and $X=001$

So to find S We need n queries $O(n)$

Quantum solution:



The same circuit
for DJ

From DJ

$$|\alpha\rangle = \sum_{y=0}^{2^n-1} \frac{1}{2^n} \sum_{x=0}^{2^n-1} (-1)^{f(x) \oplus y \cdot x} |y\rangle$$

$$F(x) = S \cdot x$$

$$\begin{aligned} F(x) \oplus y \cdot x &= (S \cdot x) \oplus (y \cdot x) \\ &= (S \oplus y) \cdot x \end{aligned}$$

$$|\alpha\rangle = \sum_{y=0}^{2^n-1} \left[\frac{1}{2^n} \sum_{x=0}^{2^n-1} (-1)^{(S \oplus y) \cdot x} \right] |y\rangle$$

Coefficient of $|y\rangle$

in case $y = S$ $S \oplus y = 0$

$$\frac{1}{2^n} \sum_{x=0}^{2^n-1} (-1)^0 = \frac{1}{2^n} \cdot 2^n$$

Coefficient of $|S\rangle = 1$

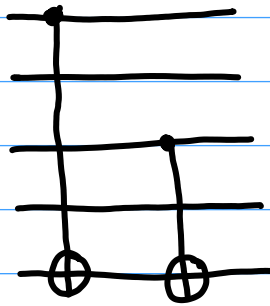
$$p(S) = |1|^2 = 1$$

so after applying BV, the output is

S With prob = 1 using only one query
 $O(1)$

eg.

U_f if $S = 1010$



This is how to apply
 $F(x)$ with S